

POC REQUIREMENTS DOCUMENT

Project Name: **Capacity & Utilization Intelligence Agent (CUIA)**

Project Type: Proof of Concept

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Objective:

This Proof of Concept validates how deterministic workforce analytics, combined with an AI-powered Copilot, can transform engineering workforce data into actionable insights for Delivery Managers and Leadership. The POC focuses on demonstrating the core analytics engine, interactive dashboards, and AI-assisted explanations using a simulated engineering organization, without the complexity of enterprise integrations or production infrastructure.

Tech Stack:

- Frontend: React, Tailwind CSS, Recharts
- Backend: FastAPI, Python
- Analytics: Pandas, NumPy (Deterministic Analytics)
- AI: LangGraph, Google Gemini

In Scope:

- Simulated engineering dataset representing Delivery Managers and Engineers.
- Deterministic workforce analytics (utilization, workload, productivity, estimation accuracy, and team health).
- Rule-based recommendation engine based on calculated analytics.
- Leadership and Delivery Manager dashboards with interactive visualizations.
- AI Copilot for natural language explanations and insights using pre-calculated analytics.
- Modular application architecture suitable for future enhancements.

Out of Scope:

- Authentication and Role-Based Access Control (RBAC).
- Live integrations with Jira or other external systems.
- CSV uploads, leave management, and skills management.
- User administration, configuration management, and audit logging.
- Background synchronization, scheduling, and notification services.
- Multi-tenancy, high availability, CI/CD pipelines, and production deployment.
- Mobile applications and enterprise security features.
- AI-generated analytics, autonomous decision-making, or data modification.

FUNCTIONAL REQUIREMENTS:

- **FR-1:** Load and manage a predefined engineering dataset representing a simulated organization.
- **FR-2:** Calculate workforce analytics deterministically, including utilization, workload, productivity, estimation accuracy, and team health.
- **FR-3:** Generate rule-based recommendations based on calculated analytics.
- **FR-4:** Provide a Leadership Dashboard with organization-wide insights, KPIs, and team comparisons.
- **FR-5:** Provide a Delivery Manager Dashboard with team-level analytics and engineer-level metrics.
- **FR-6:** Visualize workforce insights using interactive charts, KPI cards, and summary views.
- **FR-7:** Provide an AI Copilot that answers natural language questions using only pre-calculated analytics.
- **FR-8:** Ensure the AI Copilot explains analytics and recommendations without performing calculations or modifying data.
- **FR-9:** Allow switching between Leadership and Delivery Manager views for demonstration purposes.

NON-FUNCTIONAL REQUIREMENTS:

- **Simplicity:** The solution should prioritize rapid development, clarity, and demonstration value over enterprise-grade implementation.
- **Deterministic Analytics:** All workforce metrics and recommendations must be generated through deterministic Python business logic.
- **AI Boundaries:** The AI Copilot acts solely as an explanation and reasoning layer and must never calculate metrics or modify business data.
- **Performance:** Dashboards should provide fast responses using precomputed analytics suitable for live demonstrations.
- **Architecture:** The application should follow a clean modular architecture with clear separation between the Analytics Engine, Recommendation Engine, AI Copilot, and Dashboard modules, enabling future expansion without major redesign.